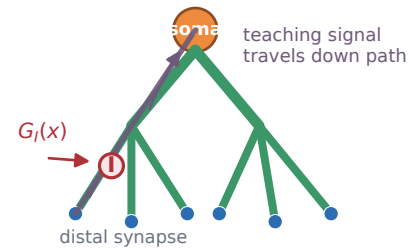


A Inhibition gates path gain

$G_I(x)$ lowers R^{tot} and attenuates all upstream paths

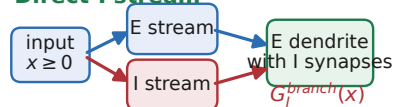


$$\alpha_n(x) = \prod_{(i \rightarrow k) \in \text{path}(n \rightarrow 0)} R_k^{tot}(x) g_{i \rightarrow k}^{den}$$
$$\partial \log \alpha_n / \partial G_k^I = -R_k^{tot} \text{ on the path}$$

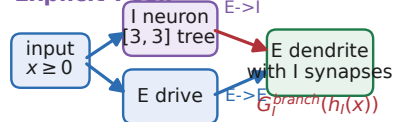
B Two inhibitory input modes

Both routes end as nonnegative I-to-E conductance on the dendrite

Direct-I stream

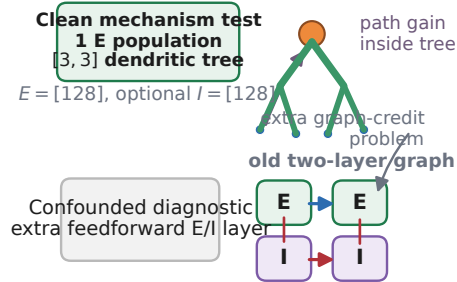


Explicit-I cell



C One dendritic layer probe

Path gain needs a branched tree, not two feedforward layers



D Probe result

